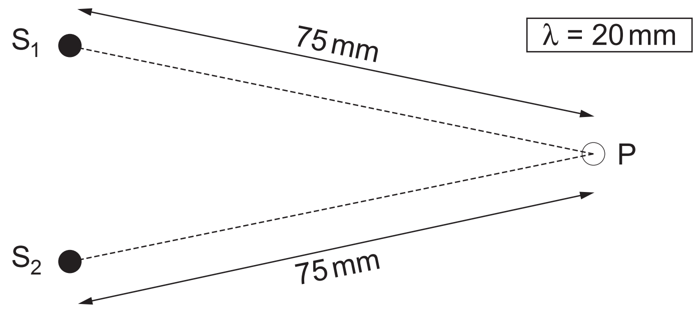
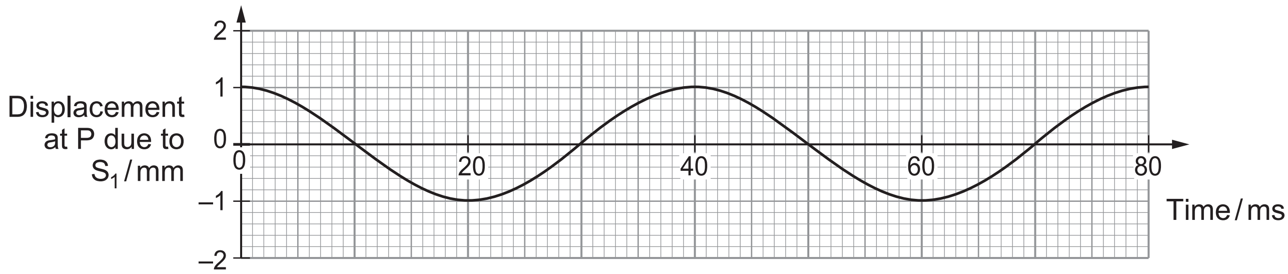
Examiner only

5. (a) The diagram shows where two metal spheres, S_1 and S_2, touch the surface of water (in a tray).

VIEW FROM ABOVE



S_2 remains stationary while S_1 is made to vibrate up and down, so waves spread outwards from it across the water. The wavelength of the waves is 20 mm. A displacement-time graph is given for point P.



(i) Calculate the wave speed. [2]

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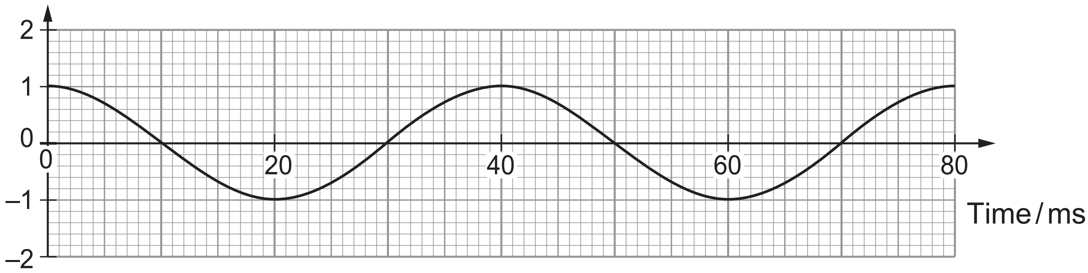
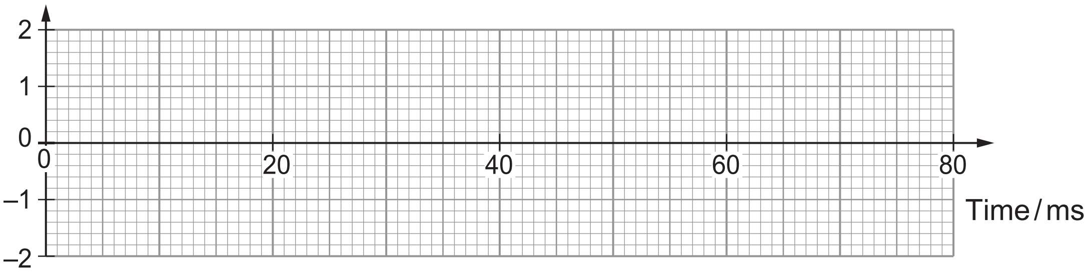
(ii) S_1 and S_2 are now made to vibrate **in phase**, to act as wave sources of equal amplitude. Carefully sketch, on the lower two grids **opposite**, displacement-time graphs for point P, due to:

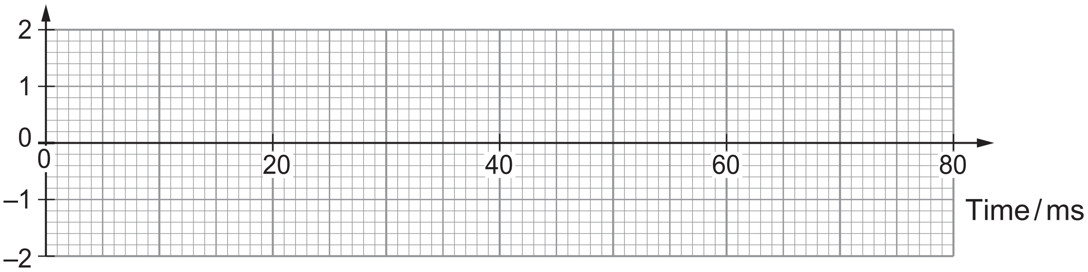
I. waves from S_2; [1]

II. the superposition of waves from S_1 and S_2. [1]

Examiner only

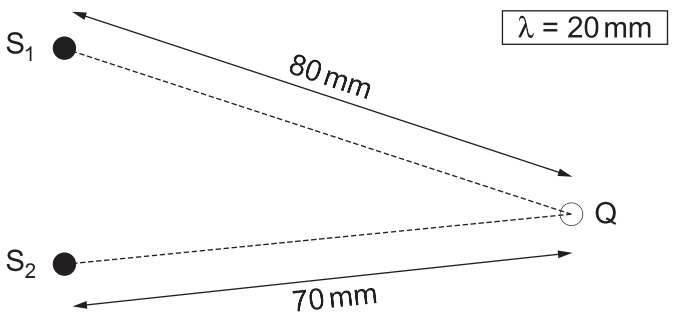






(iii) A student suggests that, with S_1 and S_2 vibrating as before, the displacement at Q (see diagram below) will be zero all the time.

VIEW FROM ABOVE



Evaluate whether or not she is correct, explaining your reasoning. [3]

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Examiner only

(b) Light of wavelength 590 nm is incident normally on a plane diffraction grating. The centres of the grating's slits are 2 000 nm apart.

(i) Calculate the angle to the normal at which **third** order beams emerge. [2]

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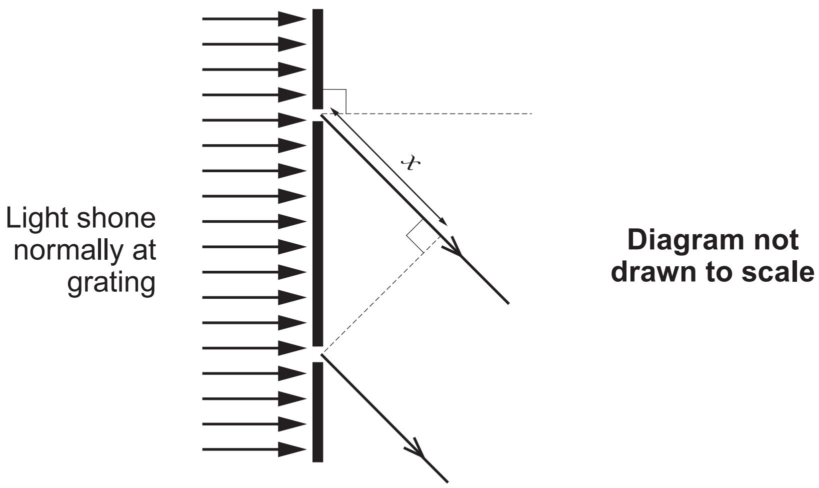
(ii) The diagram shows light emerging **at this angle** from two adjacent slits in the grating.



State the value of x. [1]

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